Along-Track Dual-Satellite D-InSAR for Multidimensional Deformation and Troposphere Joint Measurement: Configuration Optimization and Performance Analysis

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***Abstract*—Differential synthetic aperture radar interferometry (D-InSAR) has been extensively applied to deformation monitoring. When deformation can be negligible, the differential interferometric phase mainly reflects tropospheric refractivity variations, enabling tropospheric parameters retrieval. However, conventional single-satellite D-InSAR provides only line-of-sight (LOS) deformation and integrated differential tropospheric refractivity (DTR). Multiangle observations can overcome this limitation, allowing multidimensional deformation and DTR measurement. Nevertheless, observation geometry involves inherent tradeoffs among imaging resolution, deformation retrieval accuracy, and DTR tomography performance, making the system configuration optimization essential. This article investigates an along-track dual-satellite D-InSAR (ATDS-DInSAR) system as a simple yet efficient multiangle configuration for 2-D deformation and 3-D DTR joint measurement. The configuration constraints from imaging, deformation retrieval, and DTR tomography are analyzed, and a configuration optimization method for multidimensional deformation and DTR joint measurement is proposed. Simulations show that the optimized configuration improves both deformation retrieval and DTR tomography accuracy by more than 50%. In-orbit experiments with SuperView Neo-2 further validate the method, achieving millimeter-scale cross-track and centimeter-scale along-track deformation retrieval accuracy, with the 3-D DTR tomography consistent with radiosonde, Global Forecast System (GFS), and European Centre for Medium-Range Weather Forecasts (ECMWF).**

***Index Terms*—Along-track dual-satellite system, configuration optimization, differential synthetic aperture radar interferometry, multidimensional deformation, troposphere joint measurement.**

I. Introduction

DIFFERENTIAL synthetic aperture radar interferometry (D-InSAR) enables high-precision monitoring of deformations induced by earthquakes, landslides, and other geohazard events [1], [2]. Its capability is particularly enhanced when implemented on spaceborne platforms, owing to their high resolution, wide-area coverage, and all-weather observation capability [3], [4], [5]. Specifically, two repeat-pass SAR images of the same area, acquired at different times, are employed to generate an interferogram. After removing the flat-Earth, topographic, tropospheric delay, as well as other error phases, and implementing phase unwrapping, the remaining differential interferometric phase can be used to retrieve deformation [6], [7], [8].

In conventional D-InSAR, tropospheric phase is a dominant error source that degrades deformation retrieval accuracy [9], and must therefore be compensated prior to deformation retrieval. If the deformation can be negligible, the differential interferometric phase mainly reflects tropospheric refractivity variations between two repeat-pass SAR acquisition epochs, which can be further utilized for differential tropospheric refractivity (DTR) measurement and other tropospheric parameters retrieval [10], [11]. However, due to the inherent limitations of SAR imaging geometry, conventional D-InSAR provides only 1-D deformation and integrated DTR along the line-of-sight (LOS) direction. Moreover, most satellites operate in sun-synchronous orbit, which makes deformation retrieval insensitive to north–south components and thereby limits their effectiveness for hazard assessment [12].

Multiangle spaceborne SAR systems overcome this limitation by enabling observations of the same scene from different viewing angles, which allows multidimensional deformation and DTR measurement [13]. Three primary implementations can be identified. The first is a distributed system with long baselines [Fig. 1(a)], where multiple satellites at different

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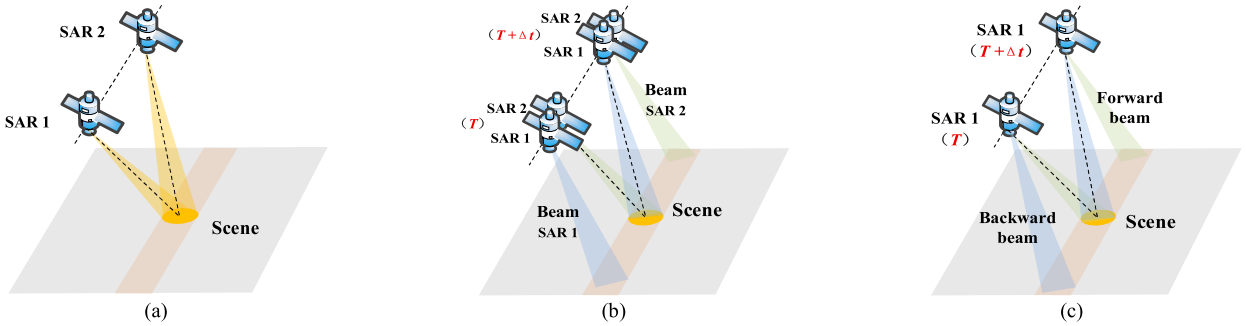


Fig. 1. Sketch map of three multiangle SAR Systems. (a) Distributed system with long baselines. (b) Distributed system with short baselines. (c) Single-satellite system with multibeam.

orbital positions acquire simultaneous multiangle observations. The second is also a distributed system but with short baselines [Fig. 1(b)], in which multiple satellites steer their beams toward different directions, enabling near-simultaneous multiangle observations during orbital motion. The third is a single-satellite system with multibeam [Fig. 1(c)], which achieves near-simultaneous multiangle observations through beam switching during orbital motion. In the latter two systems, the extremely short observation time delay Δt ensures that both surface and atmosphere remain quasi-stable, thereby providing consistent deformation and refractivity measurements [14].

Several multiangle spaceborne SAR systems have been developed worldwide. Germany's TerraSAR-X employs a bi-directional SAR (BiDi-SAR) mode, which enables forward- and backward-looking beams for near-simultaneous multiangle observations [15], [16]. ESA's upcoming "Harmony" mission, scheduled for launch in 2029, will operate in coordination with Sentinel-1 in a one-transmitter/three-receiver configuration to achieve simultaneous multiangle observations [17]. In China, the SuperView Neo-2 dual-satellite constellation, launched in November 2024, can operate in an along-track long-baseline configuration in its experimental stage, which achieves simultaneous dual-angle observation with transmit-once, dual-receive mode [18], [19].

Multiangle D-InSAR systems have been widely studied for multidimensional deformation retrieval, configuration design, and atmospheric effect compensation. For example, Wang et al. [20] proposed a multidimensional deformation retrieval method and optimal satellite configuration for global navigation satellite system (GNSS)-based InSAR. Adam introduced an atmospheric effect compensation method that exploits high-resolution numerical weather prediction (NWP) data for interferogram correction [21].

For atmospheric measurement, most existing studies have primarily focused on 1-D atmospheric parameter retrieval utilizing interferograms. For example, Mateus et al. [22] proposed a precipitable water vapor (PWV) estimation method and validated the results with Sentinel-1A data. Pichelli et al. [23] retrieved integrated water vapor and assimilated it into NWP models. More recently, Li et al. [24] developed a differential tropospheric tomography method based on a spaceborne synchronous multiangle D-InSAR system, and verified its effectiveness through simulations.

Most existing spaceborne D-InSAR systems are designed for topographic mapping and deformation monitoring, with atmospheric effects treated only as errors to be compensated [25]. Moreover, deformation is absent at most times, leading to extremely low data utilization and severely limiting the effectiveness of InSAR [26].

Therefore, a system capable of deformation and troposphere joint measurement within the same configuration is of great significance. In regions with pronounced deformation, accurate deformation can be obtained after atmospheric compensation. When it comes to areas without evident deformation, DTR can be measured to enhance the accuracy of atmospheric observation and weather forecasting [27]. Such a joint strategy avoids the complex deployments of multiple systems, thereby enhancing both efficiency and scientific value.

However, joint measurement for multidimensional deformation and DTR remains challenging in two main aspects. First, satellite orbits and observation configurations must be designed to balance the respective precisions of deformation retrieval and DTR tomography [28]. Second, a fundamental tradeoff exists in the observation geometry: large viewing angle difference improves the measurement accuracy of multidimensional deformation, but it does not necessarily guarantee higher accuracy of DTR measurement because an excessively large difference will reduce the number of retrievable voxels [29]. To date, effective methods for configuration optimization of multidimensional deformation and DTR joint measurement are still lacking.

Among multiangle systems, along-track dual-satellite D-InSAR (ATDS-DInSAR) systems represent one of the simplest yet effective solutions, offering several key advantages. First, compared with three- or multisatellite systems, ATDS-DInSAR systems are easier to implement. The reduced complexity of orbital control, intersatellite communication, and data synchronization makes them particularly suitable for disaster emergency monitoring and rapid response [30], [31]. Second, unlike conventional single-satellite D-InSAR that measures only LOS deformation, ATDS-DInSAR systems enable 2-D deformation retrieval in both along-track and cross-track directions, thereby effectively addressing the critical gap in north–south deformation monitoring for disaster prevention

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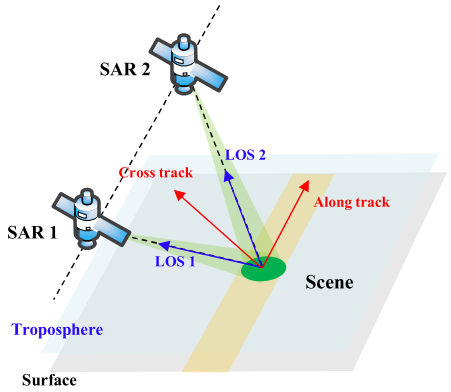


Fig. 2. Sketch map of ATDS-DInSAR system.

and mitigation [32]. Moreover, the spatial diversity provided by dual-view observations supports DTR tomography, thus realizing multidimensional deformation and DTR joint measurement at relatively low cost.

This article investigates multidimensional deformation and DTR joint measurement based on an ATDS-DInSAR system, and proposes a configuration optimization method that balances the respective precisions of deformation retrieval and DTR tomography. First, the signal and retrieval models for multidimensional deformation retrieval and DTR tomography are introduced. Next, configuration constraints, including imaging resolution, deformation retrieval accuracy, and DTR tomography performance, are analyzed, forming the basis of a configuration optimization method for multidimensional deformation and DTR joint measurement. Finally, the effectiveness of the proposed method is verified using both simulation and in-orbit experiments.

The rest of this article is structured as follows. Section II presents the methodology, including the signal and retrieval models for multidimensional deformation retrieval and DTR tomography. Section III proposes a configuration optimization method for multidimensional deformation and DTR joint measurement based on the ATDS-DInSAR system. Sections IV and V provide simulations and in-orbit experiments results, respectively, to validate the effectiveness of the proposed method. Finally, Section VI concludes this article.

II. Signal and Retrieval Model

The key to achieving multidimensional deformation and DTR joint measurement lies in the joint processing of multiangle differential interferometric phases over the same scene. As illustrated in Fig. 2, the ATDS-DInSAR system consists of two SAR satellites operating in the same orbit. By employing forward- and backward-looking beams, respectively, to observe the same scene, the system acquires multiangle imaging results, which are jointly processed at the information-level to retrieve both deformation and DTR.

After removing the flat-Earth phase and topographic phase with the assistance of an external digital elevation model (DEM), and mitigating the ionospheric phase through the split-spectrum method [33], [34], the remaining differential interferometric phase can be expressed as

$$\varphi_{int} = \varphi_{def} + \varphi_{tro} + \varphi_{noise} \tag{1}$$

where φ_{def} denotes the deformation phase during the observation period, φ_{tro} represents the tropospheric phase determined by DTR, and φ_{noise} accounts for noise-related errors.

A. Deformation Retrieval

For deformation retrieval, the tropospheric phase φ_{tro} will be compensated using accurate ray-tracing methods assisted by external meteorological data [35]. The residual phase $\boldsymbol{\varphi}'_{\mathbf{int}}$ can then be employed for 2-D deformation retrieval with two different views, which can be expressed as

$$\boldsymbol{\varphi}'_{\mathbf{int}} = \mathbf{Hd} + \boldsymbol{\varepsilon} \tag{2}$$

where $\boldsymbol{\varphi}'_{\mathbf{int}} = (\boldsymbol{\Phi}_1, \boldsymbol{\Phi}_2)^{\mathbf{T}}$ is the compensated phase vector, and $\boldsymbol{\Phi}_1$ and $\boldsymbol{\Phi}_2$ are the differential interferometric phases from two views. $\boldsymbol{\varepsilon}$ denotes the observation noise vector, and $\mathbf{d}$ is the 2-D deformation vector, which is defined as

$$\mathbf{d} = (\mathbf{d}_{\mathbf{ct}}, \mathbf{d}_{\mathbf{at}})^T = (\langle \mathbf{d}_{\mathbf{true}}, \mathbf{n}_{\mathbf{ct}} \rangle, \langle \mathbf{d}_{\mathbf{true}}, \mathbf{n}_{\mathbf{at}} \rangle)^{\mathrm{T}} \tag{3}$$

where $\mathbf{d}_{\mathrm{ct}}$ and $\mathbf{d}_{\mathrm{at}}$ denote the cross-track and along-track deformation components, respectively. The cross-track direction is aligned with the bisector of the two LOS directions, while the along-track direction lies within the plane defined by the two LOS directions and is orthogonal to the cross-track direction. Here, $\mathbf{n}_{\mathbf{ct}}$ and $\mathbf{n}_{\mathbf{at}}$ denote the corresponding unit vectors of each direction, $\mathbf{d}_{\mathbf{true}}$ represents the true deformation, and the operator $\langle \cdot, \cdot \rangle$ denotes the inner product.

$\mathbf{H}$ is the 2-D deformation observation matrix, which can be expressed as

$$\mathbf{H} = -\frac{4\pi}{\lambda} \begin{bmatrix} \langle \mathbf{n}_{\mathbf{ct}}, \mathbf{n}_{\mathbf{los}_1} \rangle & \langle -\mathbf{n}_{\mathbf{at}}, \mathbf{n}_{\mathbf{los}_1} \rangle \\ \langle \mathbf{n}_{\mathbf{ct}}, \mathbf{n}_{\mathbf{los}_2} \rangle & \langle -\mathbf{n}_{\mathbf{at}}, \mathbf{n}_{\mathbf{los}_2} \rangle \end{bmatrix} \tag{4}$$

where $\mathbf{n}_{\mathbf{los}_1}$ and $\mathbf{n}_{\mathbf{los}_2}$ are the unit vectors of the two LOS directions.

The 2-D deformation estimator $\widehat{\mathbf{d}}$ can be estimated by weighted least squares, which can be expressed as

$$\widehat{\mathbf{d}} = \left(\mathbf{H}^{\mathbf{T}} \mathbf{C}^{-1}_{\boldsymbol{\varphi}'_{\mathbf{int}}} \mathbf{H}\right)^{-1} \mathbf{H}^{\mathbf{T}} \mathbf{C}^{-1}_{\boldsymbol{\varphi}'_{\mathbf{int}}} \boldsymbol{\varphi}'_{\mathbf{int}} \tag{5}$$

where $\mathbf{C}_{\boldsymbol{\varphi}'_{\mathbf{int}}} = diag[\boldsymbol{\sigma}^2_{\boldsymbol{\Phi}_1}, \boldsymbol{\sigma}^2_{\boldsymbol{\Phi}_2}]$ denotes the covariance matrix of compensated phase errors, with $\boldsymbol{\sigma}^2_{\boldsymbol{\Phi}_1}$ and $\boldsymbol{\sigma}^2_{\boldsymbol{\Phi}_2}$ representing the variances of the differential interferometric phase corresponding to two views, respectively.

B. DTR Tomography

For short observation periods without significant deformation events (e.g., volcanic eruptions or earthquakes), the deformation can be considered negligible ($\varphi_{def} \approx 0$). In this case, the remaining differential interferometric phase is primarily attributed to tropospheric refractivity variations and can thus be utilized for DTR tomography [27]. In particular, the DTR measurement characterizes the difference in the tropospheric refractivity between two repeat-pass SAR

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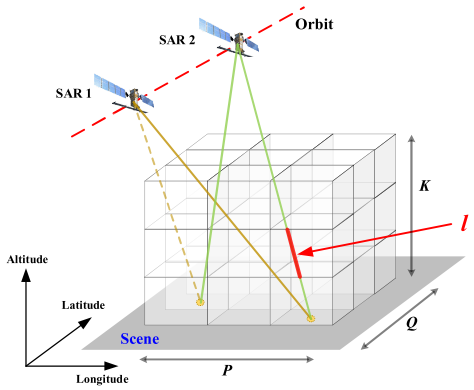


Fig. 3. Sketch map of DTR tomography measurement.

acquisition epochs, which directly reflects the atmospheric state variations.

As illustrated in Fig. 3, the observation spatial region is discretized into voxels along the horizontal and vertical directions, forming the tomographic region whose size is P, Q, and K. Only signals that fully traverse this region (solid lines) are considered valid observations, whereas those intersecting the boundaries (dashed lines) are discarded due to incomplete propagation paths.

Assuming that there are M effective observations, the corresponding differential interferometric phases are reorganized into an effective observations vector $\boldsymbol{\varphi}_{\mathbf{int}}^{''}$. The DTR tomography observation equation can then be expressed as

$$\boldsymbol{\varphi}_{\mathbf{int}}^{''} = \mathbf{L}\Delta\mathbf{N} + \varepsilon \tag{6}$$

where $\Delta\mathbf{N}$ denotes the DTR vector formed by the DTR of each voxel between two acquisition epochs. $\mathbf{L}$ is the observation matrix, which can be written as

$$\mathbf{L} = 10^{-6}\frac{4\pi}{\lambda}\begin{bmatrix} l_{1,1,1,1} & \cdots & l_{1,p,q,k} & \cdots & l_{1,P,Q,K} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ l_{M,1,1,1} & \cdots & l_{M,p,q,k} & \cdots & l_{M,P,Q,K} \end{bmatrix} \tag{7}$$

where each element $l_{i,p,q,k}$ denotes the subpath length of the ith observation passing through the voxel at (p, q, k). Each row of $\mathbf{L}$ corresponds to one observation, while each column represents all effective observations through a given voxel.

According to (6), the estimator $\widehat{\Delta\mathbf{N}}$ can be expressed as

$$\widehat{\Delta\mathbf{N}} = \mathrm{inv}(\mathbf{L})\,\boldsymbol{\varphi}_{\mathbf{int}}^{''} \tag{8}$$

where inv($\mathbf{L}$) denotes the generalized inverse of $\mathbf{L}$, since $\mathbf{L}$ is generally not a square matrix.

Due to the limited observation geometry of the system, the observation matrix $\mathbf{L}$ is sparse, leading to an underdetermined system. It will result in large errors using the least squares method to estimate $\Delta\mathbf{N}$ directly [24].

To address this issue, this study employs a DTR tomography method that combines precise ray-tracing for dimensionality reduction with singular value decomposition (SVD) to improve reconstruction stability [36], [37], [38].

First, the dimensionality reduction is conducted on the observation matrix $\mathbf{L}$. Based on the ray-tracing, some voxels that lack sufficient observations are removed, thus alleviating rank deficiency.

Then, the dimension-reduced observation matrix $\mathbf{L}'$ is decomposed using SVD, which can be expressed as

$$\mathbf{L}' = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathbf{T}} \tag{9}$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices, and $\boldsymbol{\Sigma}$ is a diagonal matrix containing the singular values of $\mathbf{L}'$.

The 3-D DTR estimator can be expressed as

$$\widehat{\Delta\mathbf{N}} = \mathbf{V}\boldsymbol{\Sigma}^{-1}\mathbf{U}^{\mathbf{T}}\boldsymbol{\varphi}_{\mathbf{int}}^{''}. \tag{10}$$

Finally, the estimated $\widehat{\Delta\mathbf{N}}$ is mapped back to the 3-D voxel, thereby reconstructing the spatial distribution of 3-D DTR.

III. Configuration Optimization Method

As discussed in Section II, different orbital configurations and observation geometries directly affect the performance of the measurements of the system, with inherent tradeoffs. For instance, enlarging the spatial observation angle enhances deformation retrieval precision and imaging resolution but compromises the DTR tomographic accuracy because of the reduced number of retrievable voxels. Therefore, a configuration optimization method is required to jointly design system parameters and balance the performance of multidimensional deformation and DTR measurement.

Section III solves this issue. First, configuration constraints are analyzed, including imaging resolution, deformation retrieval, and DTR tomography. Then, a system configuration optimization method based on the ATDS-DInSAR system is proposed for multidimensional deformation and DTR joint measurement.

A. Configuration Constraints

Implementing multidimensional deformation and DTR joint measurement using an ATDS-DInSAR system requires comprehensive consideration of three core constraints: imaging resolution, deformation retrieval, and DTR tomography. The following part will discuss these three types of constraints in detail.

1) Imaging Resolution Constraint: To reduce system costs, ATDS-DInSAR systems typically operate in a single-transmit dual-receive mode. Different configurations directly affect the ground imaging resolution of the auxiliary satellite.

The generalized ambiguity function (GAF) is commonly employed to evaluate the 2-D imaging resolution, which can be decomposed into range-equivalent resolution ρ_1 and Doppler-equivalent resolution ρ_2 [39]. Based on the bistatic 2-D resolutions ρ_1 and ρ_2, the worst case bistatic ground resolution is defined as

$$\rho_{\max} = \sqrt{\frac{\rho_1^2 + \rho_2^2 + \sqrt{\left(\rho_1^2 + \rho_2^2\right)^2 - 4\left(\rho_1\rho_2\sin\varphi_{\mathrm{g}}\right)^2}}{2\sin^2\varphi_{\mathrm{g}}}} \tag{11}$$

where φ_{g} denotes the angle between the ground 2-D resolution vectors [40].

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2) Deformation Retrieval Constraint: The deformation constraint is primarily reflected in its retrieval precision. To evaluate the performance of multidimensional deformation retrieval precision under different views, the position dilution of precision (PDOP), widely used in GNSSs, is introduced [26]. Owing to the consistency of system design and a relatively small angular separation, potential decorrelation factors, such as Doppler effects and spatially correlated of atmospheric delays, can be reasonably assumed to have a largely common-mode impact on the two LOS directions. Thus, the phase variance of interferograms generated from different views can be assumed to be identical ($\boldsymbol{\sigma}_{\boldsymbol{\Phi}_1}^2 = \boldsymbol{\sigma}_{\boldsymbol{\Phi}_2}^2 = \boldsymbol{\sigma}_{\boldsymbol{\Phi}}^2$). According to the classical definition of PDOP [41], the PDOP of 2-D deformation measurement can be expressed as

$$\mathrm{PDOP} = \sqrt{\frac{16\pi^2\left(\sqrt{\mathbf{E}_{\hat{\mathbf{d}}}^{\mathrm{T}}}\sqrt{\mathbf{E}_{\hat{\mathbf{d}}}}\right)}{\lambda^2\boldsymbol{\sigma}_{\boldsymbol{\Phi}}^2}} \tag{12}$$

where $\boldsymbol{\sigma}_{\boldsymbol{\Phi}}^2$ denotes the interferometric phase variance.

$\mathbf{E}_{\hat{\mathbf{d}}}$ denotes the vector of the 2-D deformation estimator, which can be expressed as

$$\mathbf{E}_{\hat{\mathbf{d}}} = diag\left[\left(\mathbf{H}^{\mathrm{T}}\mathbf{C}_{\varphi_{\mathrm{int}}'}^{-1}\mathbf{H}\right)^{-1}\right] = \left(\sigma_{\mathrm{CT}}^2, \sigma_{\mathrm{AT}}^2\right)^{\mathrm{T}} \tag{13}$$

where $\boldsymbol{\sigma}_{\mathbf{CT}}^2$ and $\boldsymbol{\sigma}_{\mathbf{AT}}^2$, respectively, denote the variances of the deformation in the cross-track and along-track. $\left(\sqrt{\mathbf{E}_{\hat{\mathbf{d}}}^{\mathrm{T}}}\sqrt{\mathbf{E}_{\hat{\mathbf{d}}}}\right)$ represents the sum of the deformation variances along the two dimensions, characterizing the integrated accuracy of the 2-D deformation retrieval.

Finally, the measurement accuracy coefficient of 2-D deformation $P_{\Delta d}$ is defined as

$$P_{\Delta d} = \sqrt{\frac{16\pi^2}{\lambda^2} tr\left[\left(\mathbf{H}^{\mathrm{T}}\mathbf{H}\right)^{-1}\right]} \tag{14}$$

where $tr(\cdot)$ denotes the trace of the matrix.

3) DTR Tomography Constraints: The DTR tomography constraints include retrieval accuracy and number of retrievable voxels. Based on (6), the DTR tomography accuracy coefficient $P_{\Delta N}$ can be defined as

$$P_{\Delta N} = tr\left[\left(\mathbf{L}^{\mathrm{T}}\mathbf{C}_{\varphi_{\mathrm{int}}''}^{-1}\mathbf{L}\right)^{-1}\right] = \sum_{i=1}^{P\cdot Q\cdot K}\sigma_{\Delta N_i}^2 \tag{15}$$

where $\mathbf{C}_{\varphi_{\mathrm{int}}''}$ denotes the covariance matrix of the effective observation phase errors, and $\sigma_{\Delta N_i}^2$ represents the variance of the DTR estimator for the ith voxel, and $P \cdot Q \cdot K$ is the total number of voxels in the tomographic space.

In addition, only voxels that are fully traversed by signals can be retrieved. The proportion of uncovered voxels $R_{\Delta N}$ is defined as

$$R_{\Delta N} = 1 - \frac{V}{P \cdot Q \cdot K} \tag{16}$$

where V is the number of retrievable atmospheric voxels.

In summary, configuration constraints with imaging resolution, deformation retrieval, and DTR tomography are identified, each exerting a distinct impact on system configuration. Based on these constraints, Section III-B proposes a configuration optimization method for deformation and DTR joint measurement of the ATDS-DInSAR system.

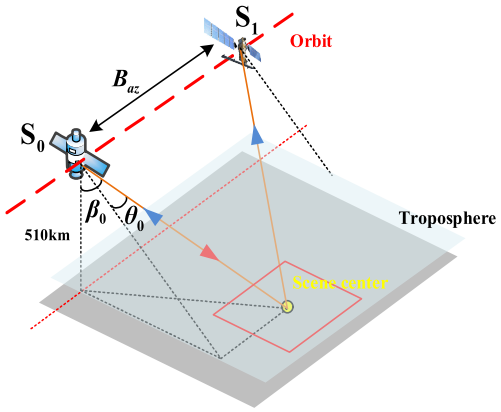


Fig. 4. Sketch map of the geometric configuration of the ATDS-DInSAR system.

B. Optimization Model Establishment and Solution

During the optimization process, parameters such as orbital altitude, frequency band, and signal bandwidth are first specified as known system parameters, since these parameters are typically fixed in a given spaceborne D-InSAR system. Accordingly, the proposed optimization method is conducted under predefined system parameters to optimize the observation configuration geometric parameters.

For a fixed orbital altitude, the observation configuration of the ATDS-DInSAR system is fully characterized by three parameters: the primary satellite's incidence angle β_0, the squint angle θ_0, and the along-track baseline B_{az} between the two satellites, as illustrated in Fig. 4.

In Fig. 4, the gray region represents the Earth's surface, the light-blue layer denotes the troposphere, the red box outlines the observation scene, and the yellow dot marks its center. The system comprises two satellites in the same orbit: the primary satellite S_0, which operates in monostatic mode with both transmission and reception capabilities, and the auxiliary satellite S_1, which functions in bistatic mode as a receive-only platform. A sufficiently long along-track baseline is maintained in the azimuth direction, which extends tropospheric coverage and improves the accuracy of 2-D deformation retrieval.

Define the system configuration geometric parameters vector $\vec{\mathbf{G}} = \{\beta_0, \theta_0, B_{az}\}$. Obviously, the imaging resolution, deformation retrieval, and DTR tomography constraints are all functions of $\vec{\mathbf{G}}$. Based on the above analysis, the final optimized objective function is constructed as

$$\begin{cases} f_1 = \min \rho_{\max}\left(\vec{\mathbf{G}}\right) \\ f_2 = \min P_{\Delta \mathrm{d}}\left(\vec{\mathbf{G}}\right) \\ f_3 = \min P_{\Delta \mathbf{N}}\left(\vec{\mathbf{G}}\right) \\ f_4 = \min R_{\Delta \mathbf{N}}\left(\vec{\mathbf{G}}\right) \end{cases} \tag{17}$$

where f_1 denotes the worst case bistatic ground imaging resolution. A smaller f_1 corresponds to higher imaging resolution. f_2 denotes the 2-D deformation retrieval accuracy coefficient. A smaller f_2 indicates higher retrieval accuracy. f_3 denotes the

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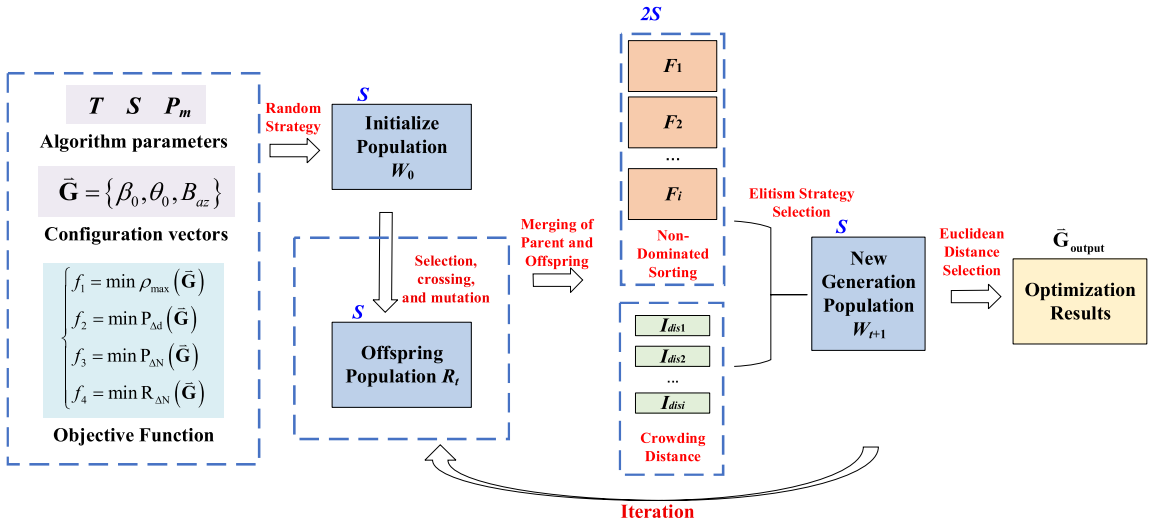


Fig. 5. Flowchart of configuration optimization method for multidimensional deformation and DTR joint measurement based on the ATDS-DInSAR system.

DTR tomography accuracy coefficient. A smaller f_3 implies higher precision in DTR tomography. f_4 denotes the proportion of uncovered voxels in the DTR tomography. A smaller f_4 means more voxels are traversed by the signals, ensuring higher completeness of DTR tomography.

Thus, the configuration optimization is formulated as a multiobjective optimization problem subject to multiple constraints [42]. To address this problem, the Non-Dominated Sorting Genetic Algorithm II (NSGA-II) is adopted [43]. The flowchart is shown in Fig. 5, and consists of the following steps.

1) Parameter Initialization: Determine the range and step size of the configuration vector $\vec{\mathbf{G}}$, as well as the population size, the number of iterations, and the mutation probability.

2) Population Initialization: Randomly generate an initial parent population W_1.

3) Fitness Evaluation: Apply genetic operators—including selection, crossover, and mutation—to generate the offspring population R. Merge the parent and offspring populations, and calculate the fitness of each individual using the objective functions in (17).

4) Nondominated Sorting: Apply fast nondominated sorting and crowding distance calculation, and use the elitism strategy to select the next generation parent population W. Repeat Step 3 iteratively to search for the optimal solution.

5) Optimal Solution Selection: After reaching the preset number of iterations, the optimization yields a set of candidate solutions (the first Pareto front). Each objective function is normalized, the Euclidean distance of each candidate to the origin is computed, and the individual with the minimum distance is selected as the optimal solution.

In Fig. 5, t denotes the current iteration, T denotes the number of iterations, p_m denotes the mutation probability, and S denotes the population size. By taking the optimal individual $\vec{\mathbf{G}}_{\mathbf{output}}$ obtained from the optimization as the system configuration parameter vector, the optimal observation configuration of the system can be determined.

TABLE I
SIMULATION PARAMETERS

Parameter		Value	
Perpendicular Baseline (m)		100	
Synthetic Aperture Time (s)		1	
Carrier Frequency (GHz)		9.6	
Orbital Altitude (km)		510	
Bandwidth (MHz)		166	
Scene Center		(106°E,29°N)	
Scene Size (km)		20×20	
Scene Resolution (m)		5×5	
Errors	Orbit Error (m)	⩽1	
	Residual Ionospheric Error	Correlation Coefficient	0.85
	Thermal Noise	*SNR* (dB)	10
	Synchronization Error	SNR_{syn} (dB)	40
		f_{syn} (Hz)	10

IV. SIMULATION AND ANALYSIS

This section validates the effectiveness of the proposed method through simulation analysis. First, the simulation parameters of the ATDS-DInSAR system are introduced. Then, the system configuration is optimized using the method described in Section III. Finally, both 2-D deformation and 3-D DTR are measured to verify the effectiveness of the proposed methods, followed by a comparative analysis before and after optimization to demonstrate the advantages of the joint configuration optimization method.

A. Simulation Setup

The simulations are performed on a typical primary–auxiliary ATDS-DInSAR system, with the geometric configuration shown in Fig. 4. The specific simulation parameters are listed in Table I. Based on the

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TABLE II
PARAMETERS OF THE DALES MODEL

Parameter	Value
Scene Size (km)	30×30
Maximal Altitude (km)	10
Horizontal Resolution (m)	400×400
Vertical Resolution (m)	100

TABLE III
PARAMETERS OF THE FOUR WEATHER SCENARIOS

	Ground Temperature (K)	Water Vapor Pressure (hPa)	Ground Pressure (hPa)
Case 1 (clear sky)	301.82	10.68	970.00
Case 2 (partly cloudy)	301.53	13.42	970.00
Case 3 (mostly cloudy)	301.33	18.99	970.00
Case 4 (rainy)	300.42	23.32	970.00

configuration vector, the satellite positions are determined, and SAR images and interferograms are generated. To mimic realistic observations, interferometric phase errors—including baseline error, residual ionospheric error, thermal noise, and synchronization error—are added.

To assess the deformation retrieval capability, a quasi-spherical 3-D deformation field is introduced into the simulation area, centered at the northwest corner of the region, with a maximum deformation of 0.2 m and an affected radius of 6 km. On this basis, tropospheric phase effects are further simulated and added. To accurately model the tropospheric delay, the Dutch atmospheric large eddy simulation (DALES) model is employed, which provides high-precision atmospheric parameter outputs in complex weather scenarios [44]. The DALES simulation setups are listed in Table II.

It is worth noting that the proposed method is designed to provide a generally applicable configuration optimization strategy, rather than deriving an optimal configuration for only a specific climate. Therefore, the climate is considered by simulating various typical climate conditions to ensure the configuration has robustness for almost all climatic regions.

Specifically, four typical weather scenarios, ranging from relatively stable to highly variable conditions, are simulated under varying ground temperature, water vapor pressure, and ground pressure conditions, which are shown in Table III.

In detail, the DALES model is utilized to simulate the tropospheric state under each meteorological condition. Using precise ray-tracing, the corresponding tropospheric delays are calculated along each path and converted into phase contributions, which are then added to the interferogram to generate the original interferogram.

TABLE IV
PARAMETER SETTINGS FOR CONFIGURATION OPTIMIZATION

Parameter	Range	Step size
Incidence Angle (°)	[10,45]	0.1
Squint Angle (°)	[0,60]	0.1
Along-track Baseline (km)	[10,100]	0.2

TABLE V
SYSTEM CONFIGURATION PARAMETERS BEFORE AND AFTER OPTIMIZATION

Parameter	$\vec{\mathbf{G}}_{\mathbf{pre}}$	$\vec{\mathbf{G}}_{\mathbf{post}}$
Incidence Angle (°)	32.3	19.3
Squint Angle (°)	43.6	27.2
Along-track Baseline (km)	17.2	52.8

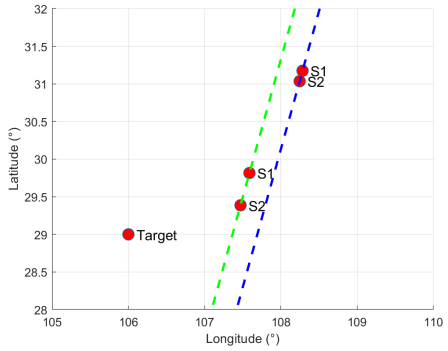


Fig. 6. 2-D top-view comparison of the configurations before and after optimization.

B. Configuration Optimization

For configuration optimization, the ranges and step sizes of the three elements in $\vec{\mathbf{G}} = \{\beta_0, \theta_0, B_{az}\}$ are set as listed in Table IV. The population size is set to 20, the mutation probability to 0.5, and the maximum number of iterations to 200. Relatively large mutation probability and iteration numbers are adopted to enhance population diversity, thereby yielding competitive optimization results.

To validate the effectiveness of the optimization, the configurations after the 10th and 200th iterations are defined as the preoptimization configuration and the postoptimization configuration, respectively. The parameter sets of two configurations are denoted as $\vec{\mathbf{G}}_{\mathbf{pre}}$ and $\vec{\mathbf{G}}_{\mathbf{post}}$, which are illustrated in Table V.

Fig. 6 presents the comparison of these two configurations from the top-view, where the blue line denotes the preoptimization configuration, while the green line corresponds to the postoptimization result.

The quantitative comparison of four objective functions before and after optimization is presented in Table VI. Results show that the objective function performance is significantly improved following configuration optimization, where the

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TABLE VI
COMPARISON OF OBJECTIVE FUNCTIONS BEFORE AND AFTER OPTIMIZATION

	Pre-optimization	Post-optimization	Improvement
f_1	15.8737	7.3930	53.4%
f_2	56.4444	14.6711	74.0%
f_3	32.6503	4.3441	86.7%
f_4	0.8159	0.5376	34.1%

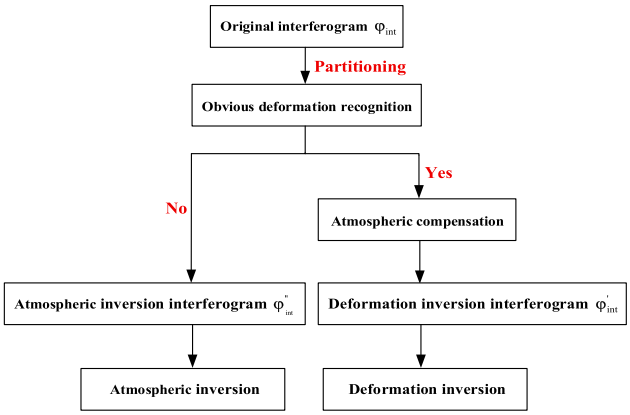


Fig. 7. Workflow for the multidimensional deformation and DTR joint measurement.

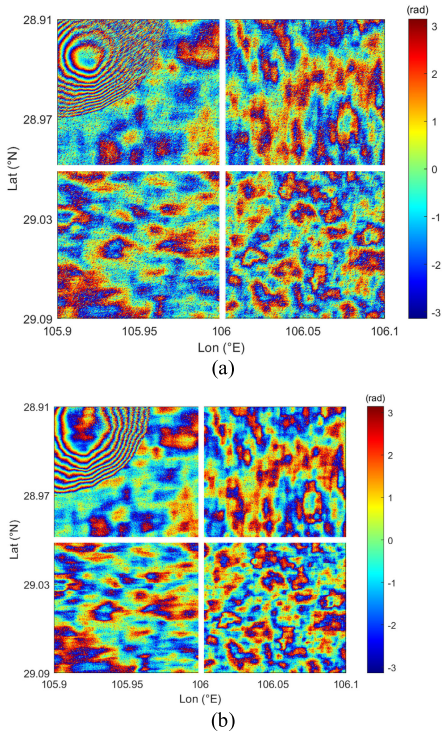


Fig. 8. Interferograms at the primary satellite's view. (a) Preoptimization. (b) Postoptimization.

improvement is defined as the percentage increase in value after optimization relative to that before optimization.

To assess the robustness of the adopted NSGA-II algorithm, additional parameter sensitivity analyses were performed. Three key parameters, including the number of iterations, population size, and mutation probability, were evaluated. The results indicate that the proposed optimization problem is not highly sensitive to the parameters, and the solutions consistently converge to the same optimized configuration $\vec{\mathbf{G}}_{\mathbf{post}}$. Detailed descriptions of the parameter sensitivity analyses and the corresponding results are provided in the Appendix.

C. Results and Analysis

Based on the original interferograms generated in Section IV-A, which contain both deformation and tropospheric phase contributions, it is necessary to separate these two components before performing the respective retrievals. The overall processing workflow is illustrated in Fig. 7.

Specifically, the interferogram is first divided into multiple sub-blocks, and significant deformation areas are identified through phase decorrelation and phase continuity analysis [45], [46].

Regarding deformation retrieval, atmospheric phases constitute a significant and non-negligible error source and therefore require compensation. Accordingly, for sub-blocks with evident deformation, atmospheric compensation is performed prior to deformation retrieval.

In contrast, for DTR tomography, the residual deformation-related phases are negligible compared with the atmospheric phases on the order of 10^0–10^1 rad. Therefore, sub-blocks without significant deformation can be directly employed for DTR tomography.

Taking the meteorological conditions of Case1 and Case4 as two observations, the 20×20-km interferograms at the primary satellite's view for the pre and postoptimized configurations are shown in Fig. 8. The region is equally divided into four 10×10-km sub-blocks. Obvious deformation, indicated by continuously interferometric fringes, is detected in the northwestern sub-block, whereas no significant deformation is observed in the other three.

1) Deformation Retrieval: According to the workflow, the northwestern sub-block is selected for deformation retrieval. First, tropospheric phase compensation is applied using coarse external NWP data. In particular, the NWP data are generated based on the DALES simulation outputs, with a spatial resolution of $4 \times 4 \times 1$ km and a zero-mean refractivity error with a standard deviation of 5 [47], [48]. The compensated interferograms are shown in Fig. 9.

It can be observed that deformation fringes are clearly visible in the northwestern region. However, due to the low resolution and errors of the external NWP data, residual tropospheric compensation phase errors remain in the interferograms. In addition, the optimized configuration yields interferograms with significantly reduced noise and clearer fringes compared with the preoptimization case.

The compensated interferograms are subsequently filtered and unwrapped to obtain deformation phases, which are used for 2-D deformation retrieval. The results are shown in Fig. 10. In the preoptimization case, unwrapping errors occur, and noise is pronounced in the along-track direction. In contrast,

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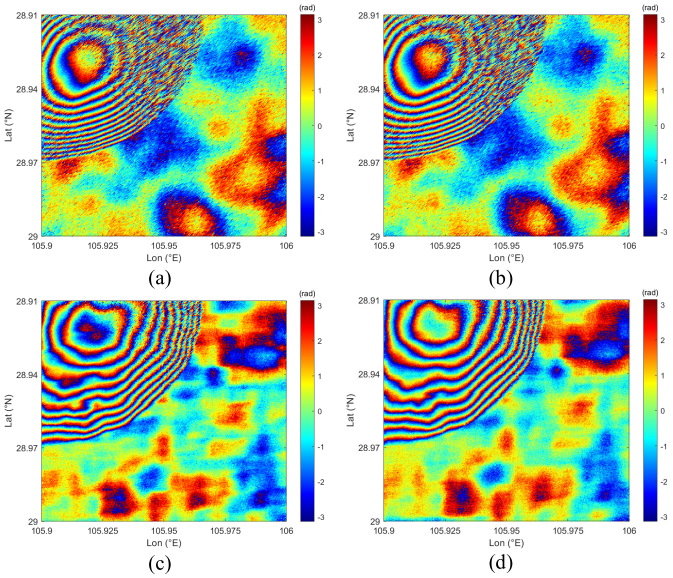


Fig. 9. Interferograms after atmospheric compensation of dual-view observation. (Left: primary satellite's view; right: auxiliary satellite's view) (a) and (b) preoptimization and (c) and (d) postoptimization.

TABLE VII

AVERAGE RMSE OF 2-D DEFORMATION RETRIEVAL BEFORE AND AFTER OPTIMIZATION (MM)

	Direction	Pre-optimization	Post-optimization	Improvement
Case1-Case2	Cross-track	10.9	4.8	55.96%
	Along-track	159.5	9.3	94.17%
Case1-Case3	Cross-track	11.4	5.1	55.26%
	Along-track	171.3	10.8	93.70%
Case1-Case4	Cross-track	12.2	5.7	53.27%
	Along-track	178.6	14.6	91.83%

the optimized configuration produces results closer to the true deformation value.

It is noteworthy that, for an ATDS-DInSAR system, the small viewing angle difference between the two LOS directions results in the cross-track deformation precision being generally an order of magnitude higher than that of the along-track direction. This conclusion is consistent with our results: the cross-track retrieval accurately captures both the magnitude and spatial pattern of the deformation, whereas the along-track retrieval identifies the deformation region and magnitude but exhibits errors at the boundaries.

To further verify robustness, 100 Monte Carlo simulations with random phase errors are conducted under all the meteorological conditions provided by the DALES outputs. The average root-mean-square error (RMSE) are shown in Table VII.

After optimization, under residual atmospheric errors, the cross-track deformation accuracy reaches millimeter-scale,

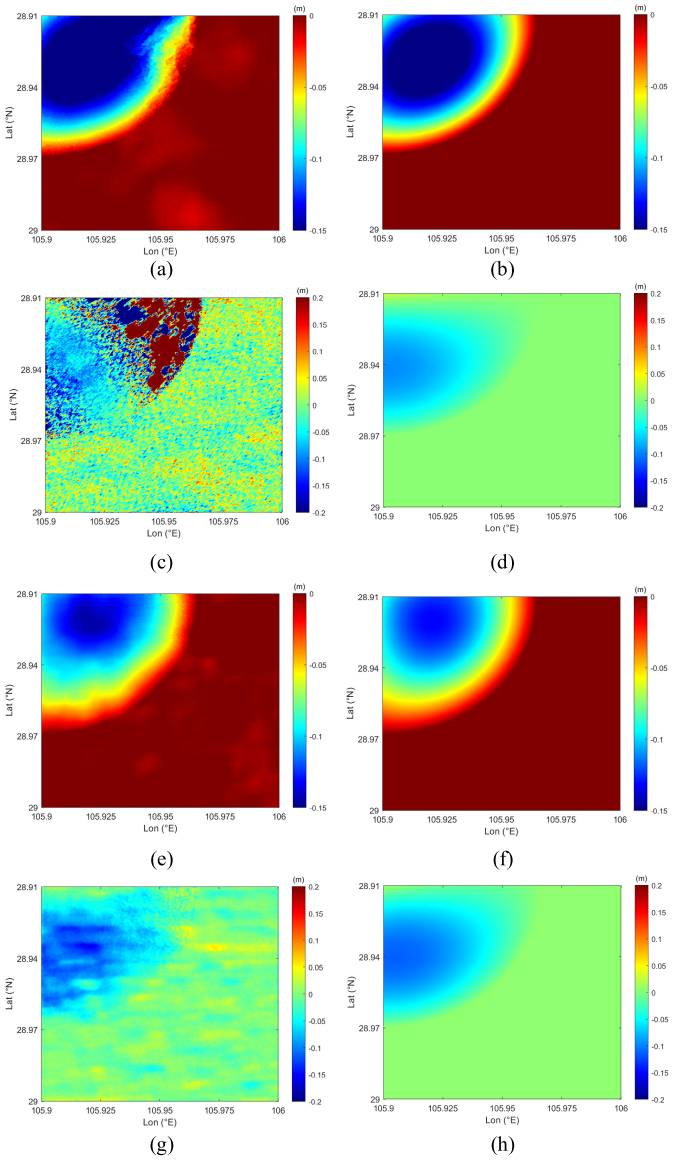


Fig. 10. 2-D surface deformation retrieval results. (Left: retrieval results; right: ground truth) (a) and (b) cross-track, preoptimization, (c) and (d) along-track, preoptimization, (e) and (f) cross-track, postoptimization, and (g) and (h) along-track, postoptimization.

while the along-track deformation accuracy reaches the centimeter scale, both showing improvements of more than 50%.

2) DTR Tomography: To further validate the effectiveness of DTR tomography after configuration optimization, a 10×10-km sub-block located in the southeast is selected. Filtering and phase unwrapping are first applied, and the resulting differential interferograms at the primary satellite's view under the three meteorological conditions are illustrated in Fig. 11.

Subsequently, the DTR tomography is implemented. Since the observation matrix in (6) is rank-deficient, a nonuniform. Voxel discretization strategy from the bottom to the top of the region is employed to reduce the number of unknowns and thereby improve the tomography accuracy. The specific parameters are summarized in Table VIII

It should be noted that the primary objective of the proposed method is to obtain high-resolution 3-D atmospheric obser-

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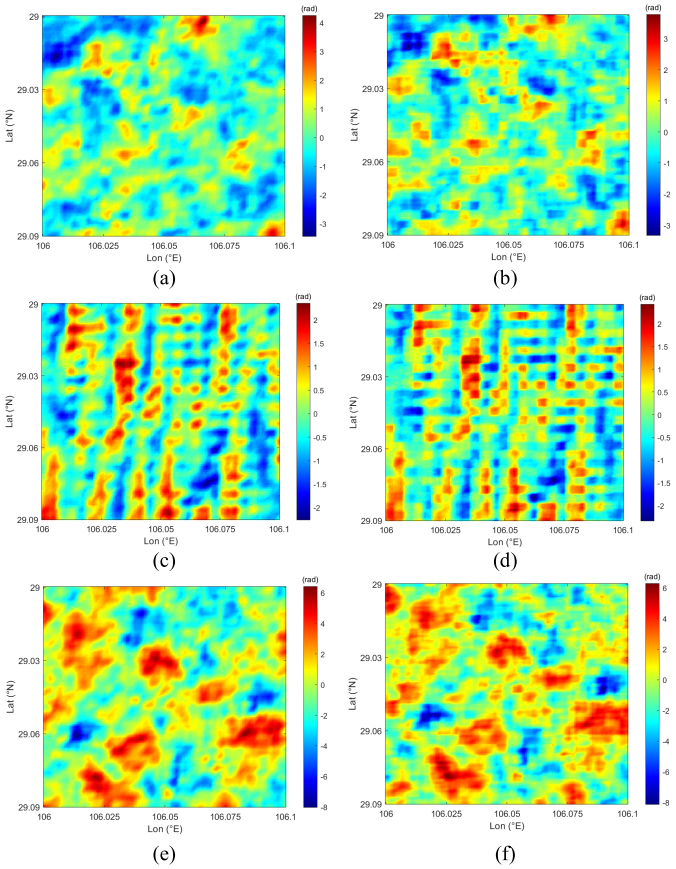


Fig. 11. Differential interferograms at the primary satellite's view under different meteorological conditions. (Left: preoptimization; right: postoptimization) (a) and (b) Case1–Case2, (c) and (d) Case1–Case3, and (e) and (f) Case1–Case4.

TABLE VIII
PARAMETERS OF THE RETRIEVAL VOXEL

Horizontal Resolution	0-4km (m)	400×400
	4-10km (m)	2000×2000
Vertical Resolution	0-4km (m)	500
	4-6km (m)	1000
	6-10km (m)	2000

vations [49], particularly to compensate for the significant limitations of conventional approaches in terms of horizontal resolution [50], [51]. Therefore, the current resolution represents a balanced tradeoff achieved by jointly considering vertical resolution, horizontal resolution, and retrieval stability.

The DTR tomography results before and after configuration optimization, along with the reference values, are illustrated in Fig. 12. It can be observed that although the whole 3-D distribution of DTR cannot be fully reconstructed due to the limitations of satellite observation geometry, the retrievable region is significantly enlarged after optimization. Moreover, under three different meteorological conditions, the optimized results exhibit smoother distributions, mitigated boundary errors, and significantly improved reconstruction of small-scale variations in the lower troposphere.

TABLE IX
COMPARISON OF DTR TOMOGRAPHIC PERFORMANCE BEFORE AND AFTER OPTIMIZATION

		Pre-optimization	Post-optimization	Improvement
Average RMSE of DTR	Case1-Case2	13.3629	5.9261	55.65%
	Case1-Case3	14.5178	5.4850	62.22%
	Case1-Case4	16.2046	6.7913	58.10%
Retrievable Voxels (total = 5100)		518	2358	355.21%

For quantitative evaluation, 100 Monte Carlo experiments with random phase errors are also performed, and the RMSE between the DTR tomography results and the reference values under different meteorological conditions is calculated for both the pre and postoptimization configurations. The average RMSE values and the number of retrievable voxels are summarized in Table IX.

The results indicate that both the retrieval accuracy and retrieval voxels are significantly improved after optimization. In Case1–Case4, the retrieval error is notably larger than in the other two conditions, which can be attributed to the strong atmospheric variability during rainy weather. This variability introduces larger quantization errors in the tomography voxels, thereby resulting in higher errors. Nonetheless, the overall performance improvements confirm the remarkable effectiveness of the optimization method, particularly under low-altitude, water vapor–dense weather conditions.

V. IN-ORBIT EXPERIMENT VALIDATION

In this section, an in-orbit experiment is conducted using the SuperView Neo-2 constellation to validate the effectiveness of the proposed method for multidimensional deformation and DTR joint measurement. First, the experimental design, including the satellite platform, observation area, and system configuration optimization, is introduced. Then, dual-view SAR images and interferograms are generated from the in-orbit data. Finally, based on the proposed approach, 2-D deformation retrieval and 3-D DTR tomography are applied, and the results are analyzed and compared with external datasets to further demonstrate the effectiveness of the method.

A. Experimental Design

The SuperView Neo-2 constellation consists of two satellites, both equipped with X-band SAR sensors capable of simultaneous dual-view observations with imaging resolutions better than 1 m, which can be operated as an ATDS-DInSAR system. A sketch map of the dual-satellite system is shown in Fig. 13. The satellites can operate in imaging, interferometry, and D-InSAR modes, with a revisit cycle of 16 days.

The autonomous in-orbit cooperative control technique has been realized for the first time worldwide, achieving orbit control precision of several hundred meters and submeter formation flying [18].

Based on the along-track formation capability of the SuperView Neo-2 constellation, the ranges of three system

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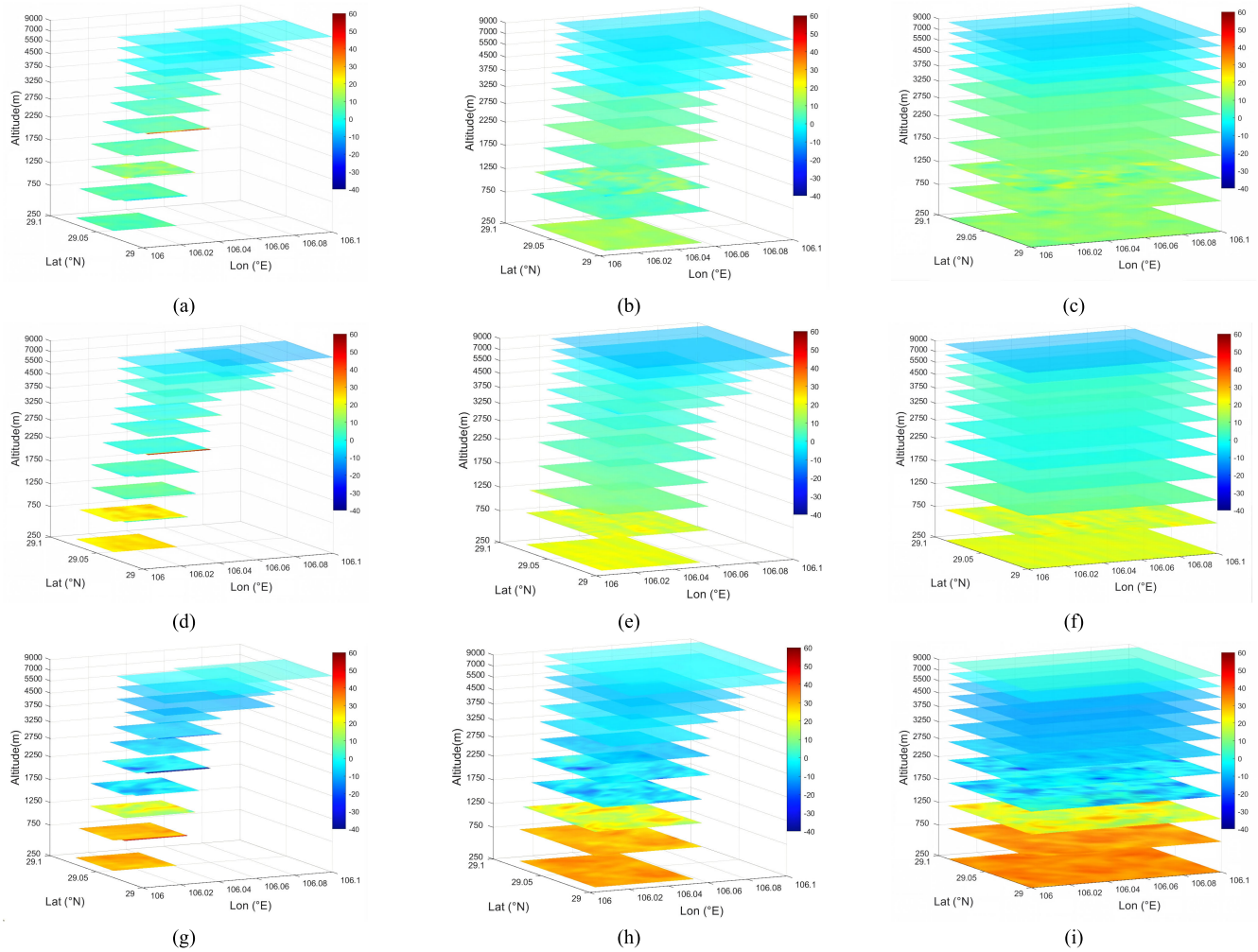


Fig. 12. 3-D DTR tomography results under different meteorological conditions. (Left: preoptimization; central: postoptimization; right: reference values) (a)–(c) Case1–Case2, (d)–(f) Case1–Case3, and (g)–(i) Case1–Case4.

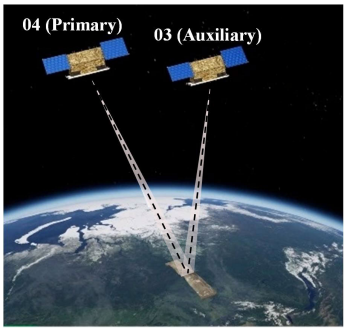


Fig. 13. Sketch map of the SuperView Neo-2 dual-satellite system.

configuration parameters were defined, and the system configuration was optimized accordingly. The parameters include the primary satellite's down incidence angle $\beta_0 \in [33.5°, 46°]$, the squint angle $\theta_0 \in [0°, 45°]$, and the along-track baseline $B_{az} \in [30, 50$ km] between the two satellites.

Using the method proposed in Section III, the optimized configuration parameter vector is denoted as $\overline{\mathbf{G}} = \{33.5°, 18.1°, 40.0$ km$\}$, and all experimental configuration parameters are summarized in Table X.

TABLE X
PARAMETERS OF THE EXPERIMENTAL CONFIGURATION

Parameter	Value
Orbital Altitude (km)	510
Incidence Angle (°)	33.5
Squint Angle (°)	18.1
Along-track Baseline (km)	40
Carrier Frequency (GHz)	9.6
Bandwidth (MHz)	166

In this configuration, the primary satellite operates in monostatic mode, while the auxiliary satellite functions in bistatic mode. The observation mode is stripmap with dual-view imaging in a transmit-once, dual-receive mode. The two acquisitions were conducted on December 4 and December 20, 2024.

B. Imaging and Interferometry

The experimental area is located in the Bayingolin Mongol Autonomous Prefecture, Xinjiang, China. SAR imaging over a

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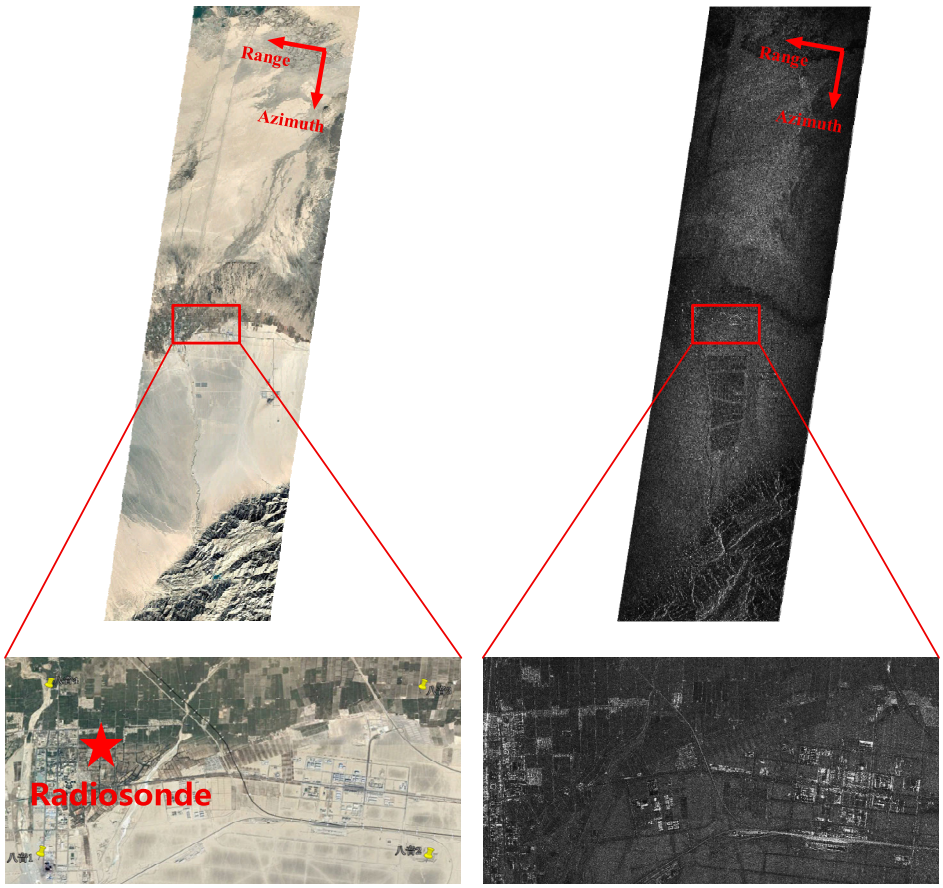


Fig. 14. Optical image and SAR imaging of the target area.

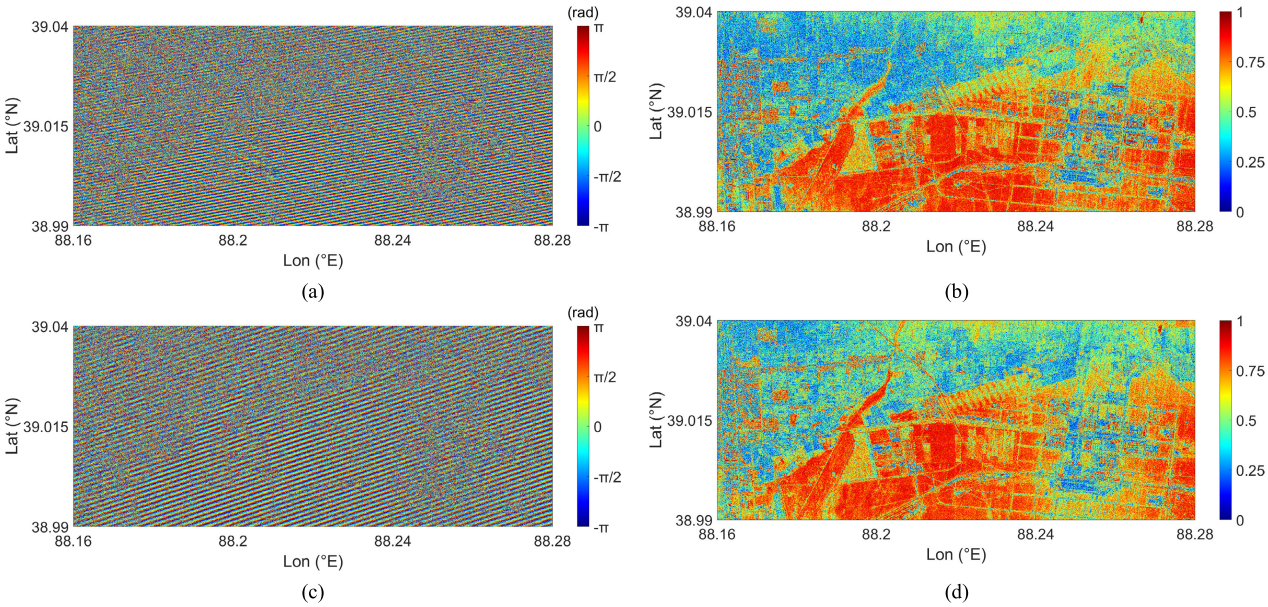


Fig. 15. Interferometric processing results. (Left: interferograms; right: coherence maps) (a) and (b) primary satellite's view and (c) and (d) auxiliary satellite's view.

100×20-km region was performed using the back-projection (BP) imaging method. Within this area, a 5×10-km subregion containing a meteorological radiosonde station was selected for high-resolution SAR imaging with a spatial resolution of 1×1 m. The optical image and the SAR image (with the primary satellite acquisition on December 4 as an example) are presented in Fig. 14, demonstrating high consistency.

The interferometric processing was performed using the December 4 acquisition as the primary image and the December 20 acquisition as the auxiliary image. As shown in Fig. 15,

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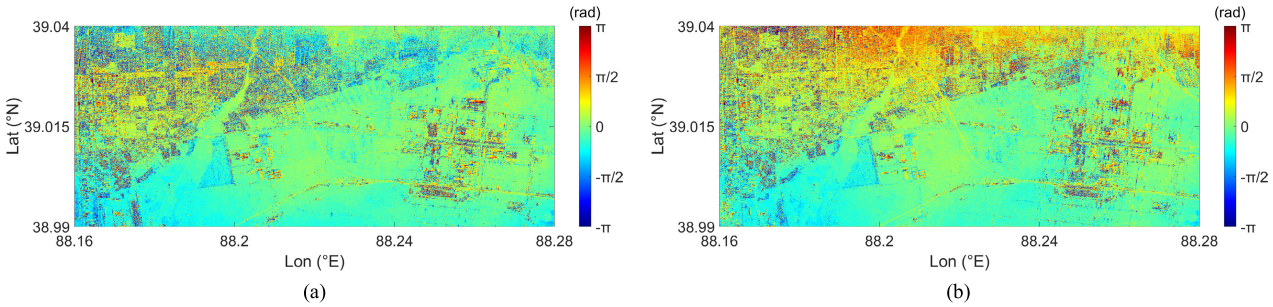


Fig. 16. Interferograms for deformation retrieval. (a) Primary satellite's view and (b) auxiliary satellite's view.

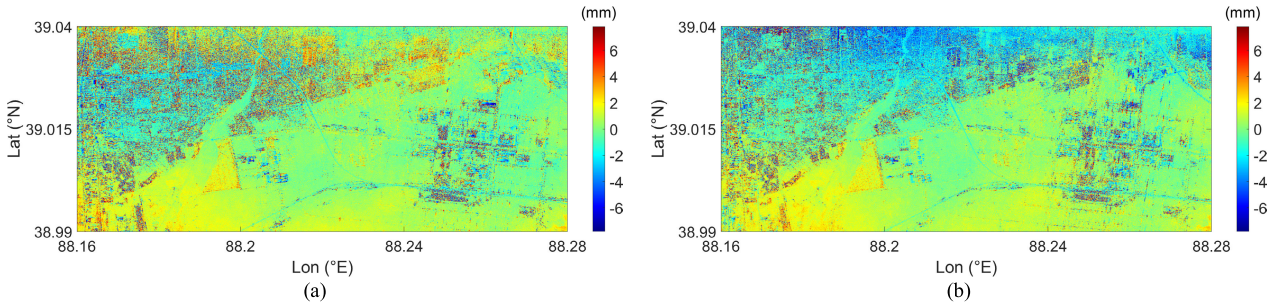


Fig. 17. Deformation results from the LOS direction. (a) Primary satellite's view and (b) auxiliary satellite's view.

TABLE XI
STATISTICS OF 1-D DEFORMATION RETRIEVAL RMSE WITH VARYING COHERENCE INTERVALS

Coherence Interval	(0.7,0.8]	(0.8,0.9]	(0.9,1.0]
LOS1 (mm)	1.6	1.5	1.3
LOS2 (mm)	1.6	1.4	1.2

TABLE XII
AVERAGE RMSE OF RETRIEVAL AND THEORETICAL

	Cross-track	Along-track
Retrieval (mm)	1.48	35.50
Theoretical (mm)	1.09	24.75

the interferograms from both views exhibit evident flat-Earth fringes, which vary with the observation geometry. Overall, the region maintains good coherence, except for reduced coherence over the northeastern oasis area.

C. Deformation and DTR Measurement

1) Deformation Retrieval: Using precise ephemerides, the flat-Earth phase and topographic phase were removed using the external DEM from SRTM-30m. Subsequently, tropospheric phase compensation was performed with ERA5 (the fifth-generation European Centre for Medium-Range Weather Forecasts (ECMWF) atmospheric reanalysis of the global climate) data. After Goldstein filtering [52], the final dual-view interferograms for deformation retrieval were obtained, as shown in Fig. 16. The phase distribution is centered near zero, with minor fluctuations observed in the northwestern region due to low coherence.

Then, LOS deformation was retrieved for both dual-view observations, and the results are shown in Fig. 17. Since no significant deformation occurred during the acquisition interval, the true deformation can be regarded as zero. As expected, the area exhibits no notable deformation, while millimeter-scale deformation errors are observed in regions with low coherence.

The RMSE of deformation retrieval was calculated in high-coherence areas (>0.7), and the results are summarized in Table XI. Both deformations for the LOS direction achieve less than 2-mm accuracy.

On this basis, 2-D deformation was retrieved in high-coherence areas (>0.7), whose results are shown in Fig. 18. The results indicate that no significant deformation occurred, with cross-track retrieval errors an order of magnitude smaller than along-track errors.

Fig. 19 shows the statistical histograms of the retrieval errors, with the retrieval and theoretical RMSE summarized in Table XII. The results indicate that the cross-track direction achieves millimeter-scale accuracy, while the along-track direction reaches centimeter-scale accuracy due to the small view angle difference. In Addition, the retrieval accuracy is slightly lower than the theoretical expectation (using an averaged $\gamma = 0.85$), which is reasonable given the presence of residual topographic and atmospheric phase errors in the interferograms.

2) DTR Tomography: Since no significant deformation occurred, the dual-view interferograms can be directly used for DTR tomography. The resulting unwrapped dual-view interferograms used for DTR tomography are shown in Fig. 20.

Due to the 33.5° incidence angle, the number of retrievable voxels and tomography accuracy are significantly lower than those achieved under the optimal configuration discussed in

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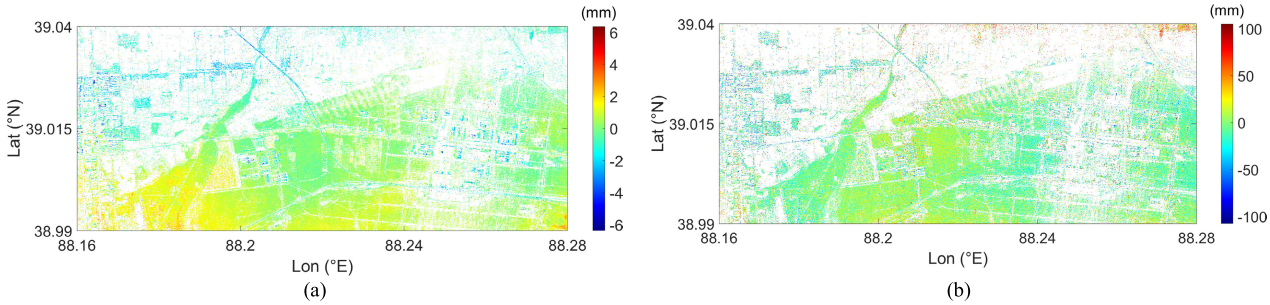


Fig. 18. 2-D deformation retrieval results in high-coherence areas. (a) Cross-track and (b) along-track.

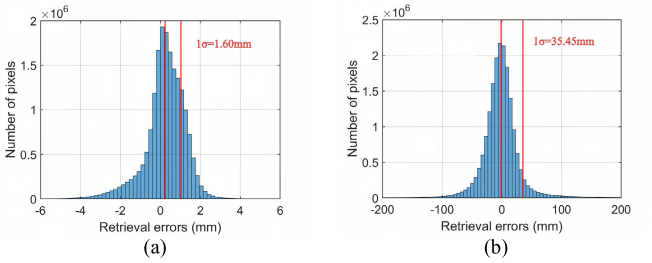


Fig. 19. Statistical histograms of the 2-D deformation retrieval errors. (a) Cross-track and (b) along-track.

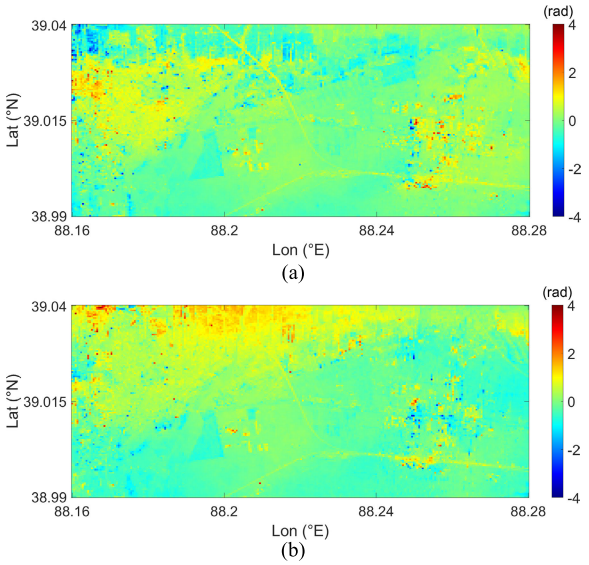


Fig. 20. Dual-view interferograms for DTR tomography. (a) Primary satellite's view and (b) auxiliary satellite's view.

TABLE XIII
RMSE STATISTICS OF THE RETRIEVAL RESULTS COMPARED WITH EXTERNAL DATASETS

Dataset	ECMWF	GFS	Radiosonde	Theoretical
RMSE	6.6053	7.6245	6.7659	5.3896

Section IV. Here, an 8×8 horizontal voxel grid with four vertical layers from 0 to 6 km was adopted. To reduce additional phase errors caused by low-coherence areas, only pixels with high coherence (>0.7) were selected as effective

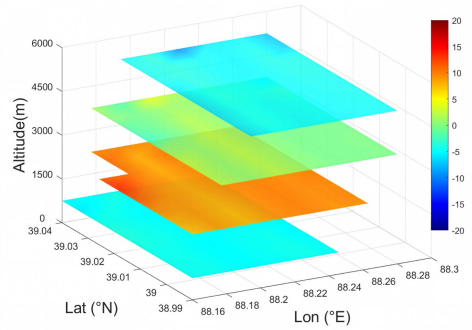


Fig. 21. 3-D DTR tomography results.

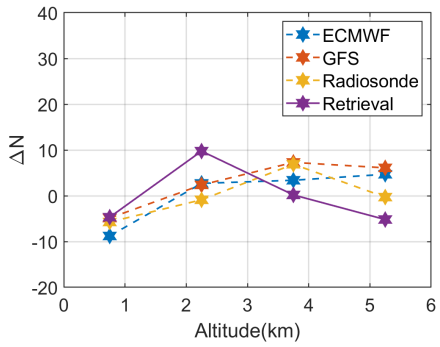


Fig. 22. Comparison of the DTR tomography results with external datasets.

observations. The results, shown in Fig. 21, indicate small fluctuations, consistent with the arid climate of the Bayingolin region [53]. In addition, some voxels at the 3-km altitude could not be retrieved because of low coherence in the northwest.

For validation, the results were compared with ECMWF, Global Forecast System (GFS), and the local radiosonde measurements. ECMWF and GFS represent the most advanced publicly accessible global numerical weather products, while radiosonde observations remain the most accurate atmospheric measurement technique, with refractivity accuracy typically better than 3 [54], [55].

Given that ECMWF and GFS have a resolution of 10-km scale and only one radiosonde station is available, the results were averaged within each vertical layer. Considering that the Bayingolin region is located in a desert area with relatively stable atmospheric conditions, this comparison strategy is reasonable and appropriate for validation.

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TABLE XIV
SENSITIVITY ANALYSIS OF THE NUMBER OF ITERATIONS

Parameter	50	100	150	200	300
Incidence Angle (°)	28.4	17.4	19.6	19.3	19.3
Squint Angle (°)	33.2	28.1	26.9	27.2	27.2
Along-track Baseline (km)	28.5	54.8	51.4	52.8	52.8

TABLE XV
SENSITIVITY ANALYSIS OF THE POPULATION SIZE

Parameter	10	20	30
Incidence Angle (°)	21.4	19.3	19.3
Squint Angle (°)	35.2	27.2	27.2
Along-track Baseline (km)	64.2	52.8	52.8

TABLE XVI
SENSITIVITY ANALYSIS OF THE MUTATION PROBABILITY

Parameter	0.1	0.2	0.5	0.7
Incidence Angle (°)	28.4	17.4	19.3	19.3
Squint Angle (°)	33.2	28.1	27.2	29.6
Along-track Baseline (km)	28.5	54.8	52.8	52.8

The comparisons are shown in Fig. 22 and statistical RMSE for retrieval results (with respect to external datasets), and theoretical RMSE are listed in Table XIII.

Overall, the results demonstrate good consistency, with errors of approximately 7. The theoretical accuracy is slightly better than the retrieval accuracy, which is reasonable considering that practical measurements are affected by quantization errors and that the external data themselves contain uncertainties.

VI. CONCLUSION

This article investigates a method for multidimensional deformation and DTR joint measurement based on an ATDS-DInSAR system, and proposes a configuration optimization method to balance their precision, respectively. Simulation experiments under different scenarios using the DALES model demonstrate that the proposed method offers significant advantages in both deformation retrieval and DTR tomography.

In terms of deformation retrieval, the optimized configuration achieves millimeter-scale cross-track deformation accuracy and centimeter-scale along-track deformation accuracy, representing improvements over 50%. For DTR tomography, the optimized configuration achieves accuracy better than 7.

Moreover, in-orbit experiments based on SuperView Neo-2 constellation further demonstrate the effectiveness of the proposed method. Using the optimized configuration, repeat-pass interferometry successfully retrieved both 2-D deformation and 3-D DTR tomography results. Results show that millimeter-scale cross-track accuracy and centimeter-scale along-track accuracy for deformation, while the DTR tomography achieved an RMSE of around seven against radiosonde, ECMWF, and GFS, confirming the reliability of the method.

Future research will focus on large-scale scenarios (thousand-kilometer swath width with subkilometer resolution) for meteorological applications. In addition, efforts will be devoted to extending the retrieval to 3-D deformation, as well as joint retrieval of additional tropospheric parameters such as water vapor pressure and cloud distribution.

APPENDIX

To assess the robustness of the adopted NSGA-II optimization algorithm, additional parameter sensitivity analyses were performed. Three key parameters, including the number of iterations, population size, and mutation probability, were evaluated. For each parameter setting, the optimization was independently analyzed, and the resulting optimized configurations were compared.

As shown in Tables XIV–XVI, the optimized solutions consistently converge to nearly identical configurations, characterized by an incidence angle of approximately 20°, a squint angle of about 28°, and an along-track baseline of around 53 km. This convergence behavior remains stable across a wide range of parameter values, including iteration numbers from 50 to 300, population sizes from 10 to 30, and mutation probabilities from 0.1 to 0.7. These results indicate that the proposed optimization problem exhibits good numerical stability and low sensitivity to NSGA-II parameter selection.

Considering both performance and computational efficiency, the parameters adopted in this article in this article (200 iterations, a population size of 20, and a mutation probability of 0.5) are reasonable and the optimized configuration $\vec{\mathbf{G}}_{\mathbf{output}}$ can be regarded as the optimal configuration.

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